Arabic Synonyms Index and Finder

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I would also like to thank all my respected teachers at the university who have contributed to my learning journey. Their dedication, support, and knowledge-sharing have helped me develop the skills necessary to complete this project.

This project would not have been possible without the collective effort, support, and mentorship I received from my instructors and the university.

Abstract

This project, titled "Arabic Synonym Index and Finder", is a web-based application designed to help users learn and improve their Arabic language skills by allowing them to search for Arabic words and find their synonyms. Users can enter words in Arabic or using transliteration, and the system will display relevant synonyms to enhance vocabulary and understanding.

During this study, it was observed that there are very few, if any, websites providing such a dedicated tool for managing and exploring Arabic synonyms. This made the project both unique and challenging. It offered a valuable opportunity to strengthen my software development skills, especially in designing data-driven applications and building useful educational tools.

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1. Software Project Management Plan (SPMP)

1. INTRODUCTION

1.1 Project Overview

This is a web-based application designed to help users search for Arabic words and retrieve their synonyms. The system allows users to input word typically in Arabic word and returns all related synonyms stored in the database. Admin users have dedicated access to manage the word and synonym entries, including adding new words, editing existing ones, or deleting outdated records, ensuring that the data remains accurate and up-to-date.

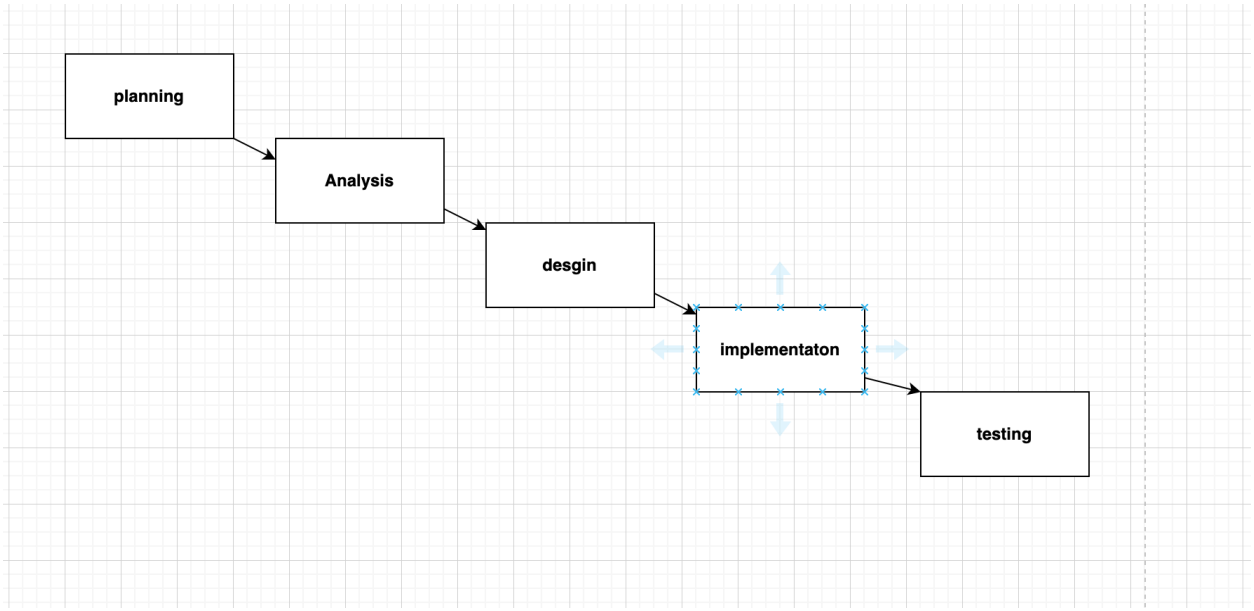
1.2 Project Deliverables

- Functional Web App
    - User Interface
    - Admin Interface
    - Synonym Search System
- Documentation including these four chapters (SPMP, SRS, SDD, STD)

2. PROJECT ORGANIZATION

2.1 Software Process Model

WaterFall Model — features are added in stages (search, view, manage, quiz).



2.2 Roles and Responsibilities

- **Developer :** Code, DB, UI, Testing (Taimoor)
- **Supervisor:** Guide and validate (Dr. Onaiza Maqbool)
- **User/Admin:** Tester, feedback provider (Taimoor)

2.3 Tools and Techniques

- **Language:** Python (Flask)
- **Database:** PostgreSQL
- **UI:** React Native
- **Tools:** vscode

3. PROJECT MANAGEMENT PLAN

3.1 Tasks

- **Planning** : Ms word,ganttpro
- **Analysis** : Draw.io ,,Ms word
- **Desgin :** Ms word , Draw.io
- **Database** : SQLite/PostgreSQL
- **Deveopment**:Vscode,github

- **Method** : structure approach

3.2 Description :

3.2.1 planning :

This includes planning of this software system “Arabic synonyms index and finder “ like defining a process model .

3.2.2 Analysis :

In analysis we define the requirement ,define the use case ,use case description.

3.2.3 design :

Design overview , requirement traceability ,user interface description and component description .

3.2.4 development :

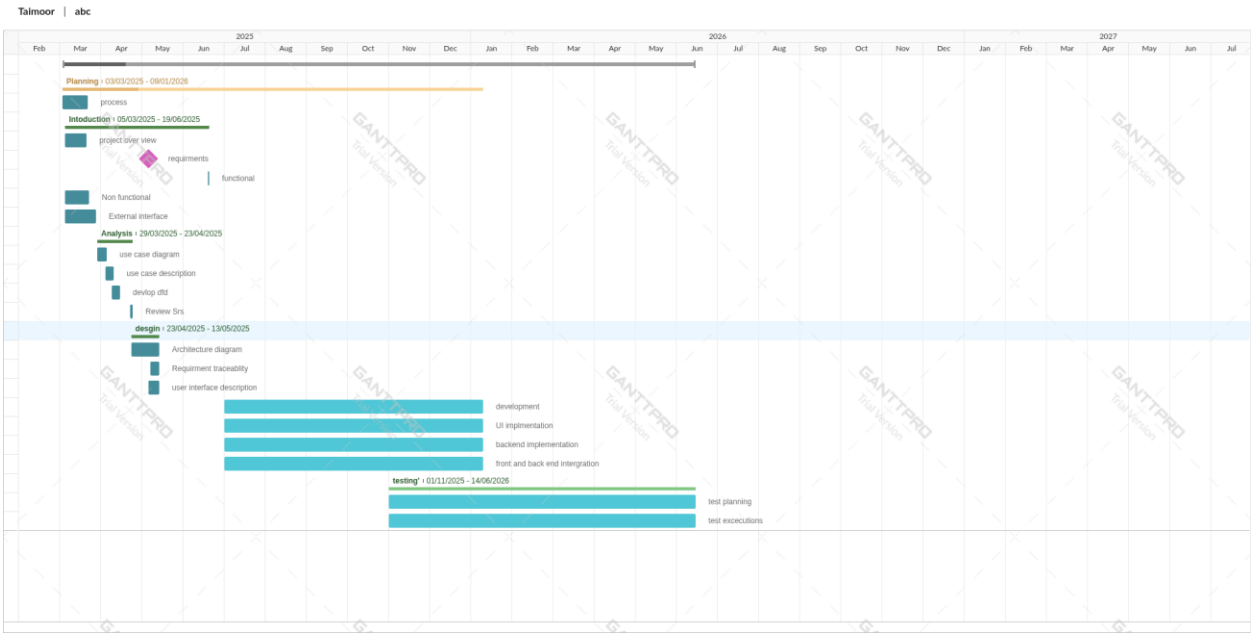
This includes the front and backend of the system using tools and frame works .

3.2.5 testing :

Test cases to check functionality on of the system

3.2 Timetable

Task name	Start date	End date
	03/03/2025	14/06/2026
Planning	**03/03/2025**	09/01/2026
process	03/03/2025	21/03/2025
Intoduction	**05/03/2025**	19/06/2025
project over view	05/03/2025	20/03/2025
requirments	06/05/2025	06/05/2025
functional	19/06/2025	19/06/2025
Non functional	05/03/2025	22/03/2025
External interface	05/03/2025	27/03/2025
Add a task | Add a milestone		
Analysis	**29/03/2025**	23/04/2025
use case diagram	29/03/2025	04/04/2025
use case description	04/04/2025	09/04/2025
devlop dfd	09/04/2025	14/04/2025
Review Srs	22/04/2025	23/04/2025
Add a task | Add a milestone		
desgin	**23/04/2025**	13/05/2025
Architecture diagram	23/04/2025	13/05/2025
Requirment traceablity	07/05/2025	13/05/2025
user interface description	06/05/2025	13/05/2025
Add a task | Add a milestone		
development	01/07/2025	09/01/2026
UI implmentation	01/07/2025	09/01/2026
backend implementation	01/07/2025	09/01/2026
front and back end intergration	01/07/2025	09/01/2026
Add a task | Add a milestone		
testing'	**01/11/2025**	14/06/2026
test planning	01/11/2025	14/06/2026
test excecutions	01/11/2025	14/06/2026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2025
2026
2027
Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul
Planning › 03/03/2025 - 09/01/2026
process
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Requirment traceablity
user interface description
development
UI implmentation
backend implementation
front and back end intergration
testing' › 01/11/2025 - 14/06/2026
test planning
test executions

2. Software Requirements Specifications (SRS)

1. INTRODUCTION

1.1 Product Overview

Arabic Synonym Index and Finder is a web app that allows users to input Arabic words and in result it find the synonyms from a list

2. REQUIREMENTS

2.1 External Interface Requirements

2.1.1 User Interfaces

- Search page (user input in Arabic)
- Synonym results page
- Admin panel for word management

2.1.2 Hardware Interfaces

- None

2.1.3 Communications Protocols

- HTTP/HTTPS

2.2 Functional Requirements

1. **Search Arabic Word**
    - Users can input an Arabic word.
    - System displays Arabic synonyms of that arabic word and also their location in the quran.

2. **View Synonyms**

    - Users can view a list of synonyms related to the searched word and also their location in the quran.

3. **Admin Word Management**

    - Admin can add new Arabic words and their synonyms.
    - Admin can edit existing words and their synonyms.
    - Admin can delete words and their synonyms.

4. **Quiz Feature**

    - System generates quizzes for users based on word/synonym matching and location in the quran.
    - Quiz results are shown after submission.

5. **User Access**

    - No login required for normal users.
    - Only login is required for Admin .

2.3 Non-Functional Requirements

1. **Performance**

    - Search results must be displayed in less than 3 seconds.

2. **Reliability**

    - The system must be available 99% of the time.

3. **Security**

    - Admin panel must be password-protected.

4. **Usability**

    - The UI must be easy to navigate for users with no technical background in short it should be user friendly and easy to use

    - Arabic fonts should be readable and properly aligned.

5. **Maintainability**

    - Codebase must be modular and well-documented.

    - Database should support easy updates to word/synonym pairs.

6. **Scalability**

    - Should be designed to scale if the list of arabic words are more or increases .

2.4 Database Requirements

- A database will be required to store Arabic words, their synonyms and location in the Quranic text

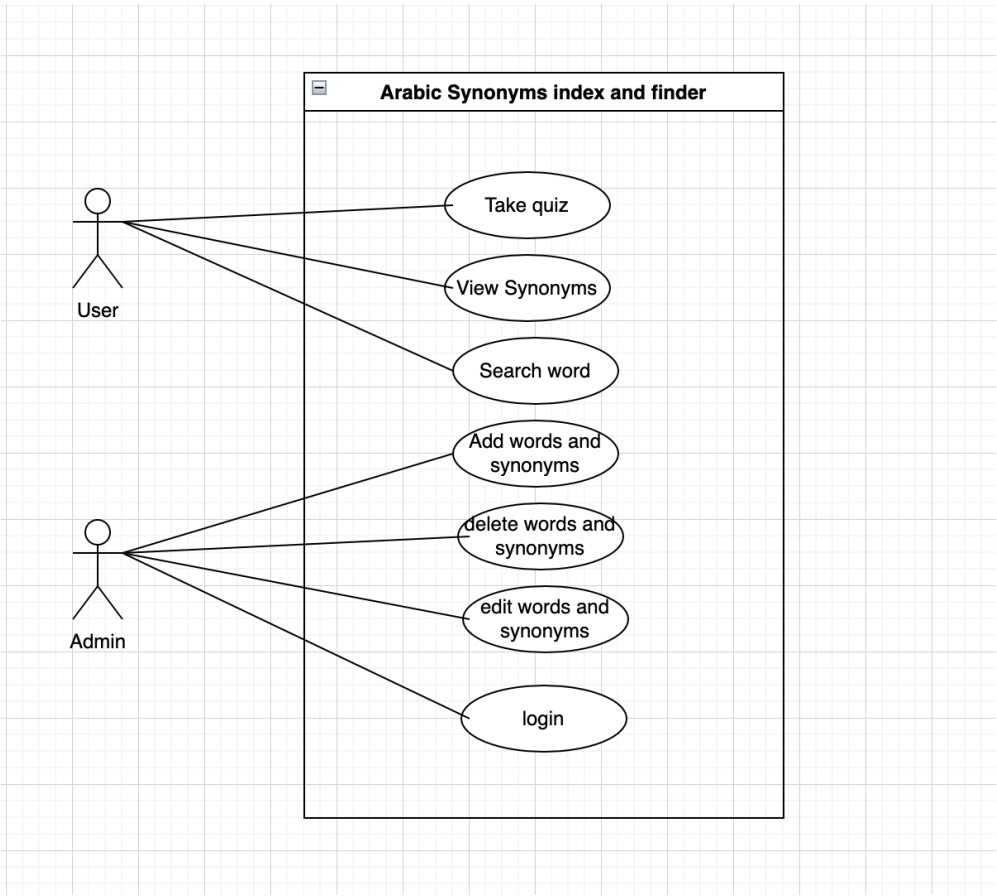
2.5 Use Case Diagram Actors:

- User
- Admin

2.5.1 Use Cases:

- Search Word
- View Synonyms
- Take Quiz
- Add word and synonyms
- Edit word and synonyms
- Delete word and synonyms

2.5.2 Use case diagram :



2.6 Use Case description:

1.

Use case	UC-01
Use case Name	Search Words
Actors	User
Description	It will allow the user to search words in Arabic and it will return the synonyms with the word which the user searched and Also identify the word and synonyms in the quran
Pre Condition	none
Post condition	The system will display the synonyms and the word which user searched .
Main Flow	1. User will search the word in Arabic or english 2. System will match the word if available in the list (database). 3. System will display the synonyms and the word which user searched.
Alternative flow	2.a System will not be able to find that particular word in the list 2.b In that case it will display " no result found"
Frequency Occurrence	HIGH

2.

Use case	UC-02
Use case Name	Take Quiz
Actors	User
Description	The user takes a quiz to test their knowledge of Arabic words and their synonyms.
Pre Condition	none
Post condition	The system displays the quiz score after submission.
Main Flow	1. User clicks on "Take Quiz". 2. System shows questions related to words and synonyms. 3. User answers and submits the quiz. 4. System checks and displays the result.
Alternative flow	2.a If no questions are available it will show message "quiz not available now"
Frequency Occurrence	Medium

3.

Use case	UC-03
Use case Name	Add Word and Synonyms
Actors	admin
Description	Admin adds a new Arabic word along with its synonyms to the list.
Pre-Condition	Admin must be logged in.
Post condition	New Arabic word and synonyms are stored in the list.
Main Flow	1. Admin clicks on "Add Word". 2. Admin enters the Arabic word and its synonyms. 3. System saves the new data.
Alternative flow	2.a If the word already exists, system shows: **"Word already exists."**
Frequency Occurrence	Medium

4.

Use case	UC-04
Use case Name	Edit Word and Synonyms
Actors	admin
Description	Admin can update an existing word or its synonyms in the database.
Pre Condition	Admin must be logged in.
Post condition	The word or synonyms are updated in the system.
Main Flow	1. Admin selects the word to edit. 2. Admin updates the information. 3. System saves the updated data.
Alternative flow	1.a If the word does not exist, show: **"Word not found."**
Frequency Occurrence	Medium

5.

Use case	UC-05
Use case Name	Delete Word and Synonyms
Actors	admin
Description	Admin deletes a word and its related synonyms from the system.
Pre Condition	Admin must be logged in.
Post condition	The selected word and its synonyms are removed.
Main Flow	1. Admin selects a word to delete. 2. System asks for confirmation. 3. Admin confirms, and the word is deleted.
Alternative flow	3.aIf deletion fails, show: **"Unable to delete the word."**
Frequency Occurrence	Medium

6.

Use case	UC-06
Use case Name	View Synonyms
Actors	User
Description	The user can view synonyms of searched Arabic word.
Pre Condition	The word must be available in the list.
Post condition	The system shows a list of Arabic synonyms related to that word.
Main Flow	1. User clicks on a word or enters a word. 2. System retrieves the synonyms from the list.. 3. System displays the list of synonyms with the searched arabic word

Alternative flow	2.a If no synonyms are found, system shows "no synonyms found."
Frequency Occurrence	Medium to High

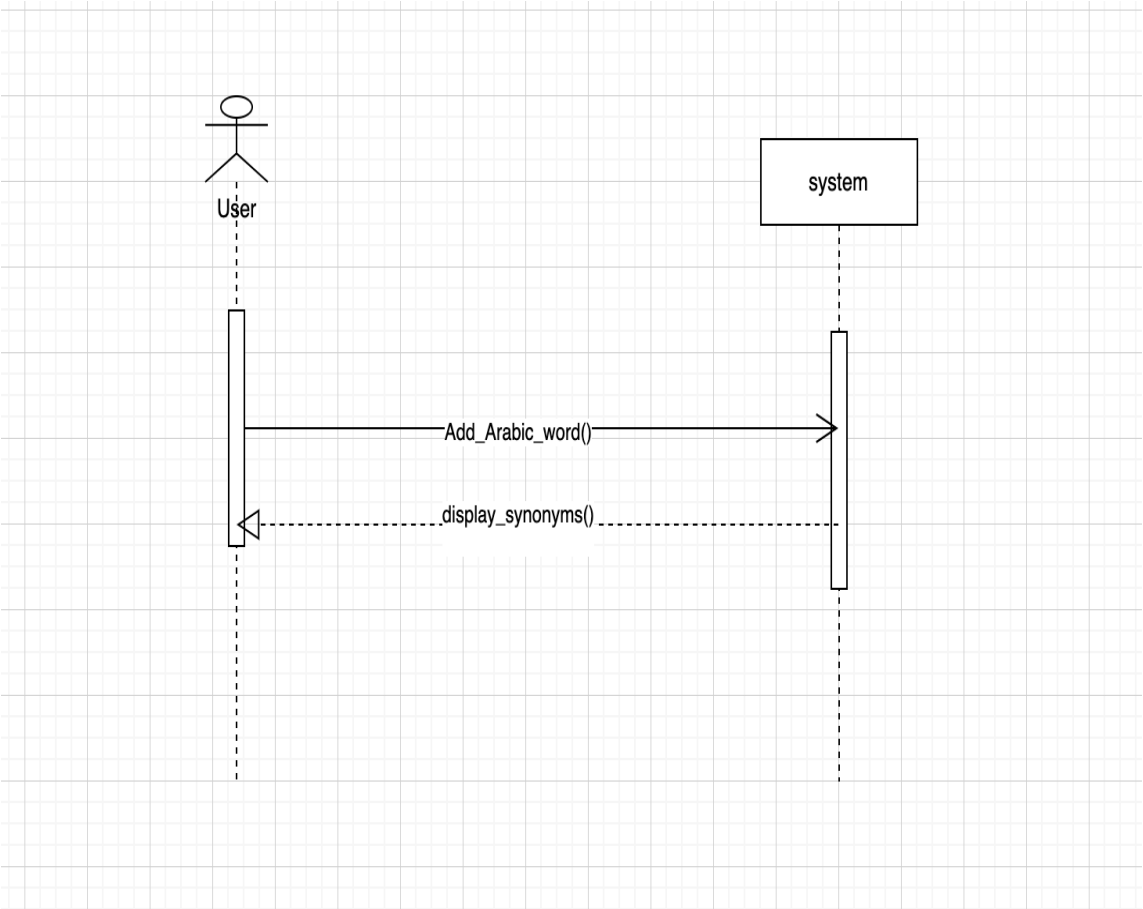
7.

Use case	UC-07
Use case Name	login
Actors	admin
Description	This use case allows the admin to log in to system to manage Arabic words and their synonyms.
Pre Condition	Admin must have valid login credentials .
Post condition	Admin is granted access to the admin panel for performing add, edit, and delete operations.
Main Flow	1. Admin navigates to the login page. 2. Admin enters username and password.

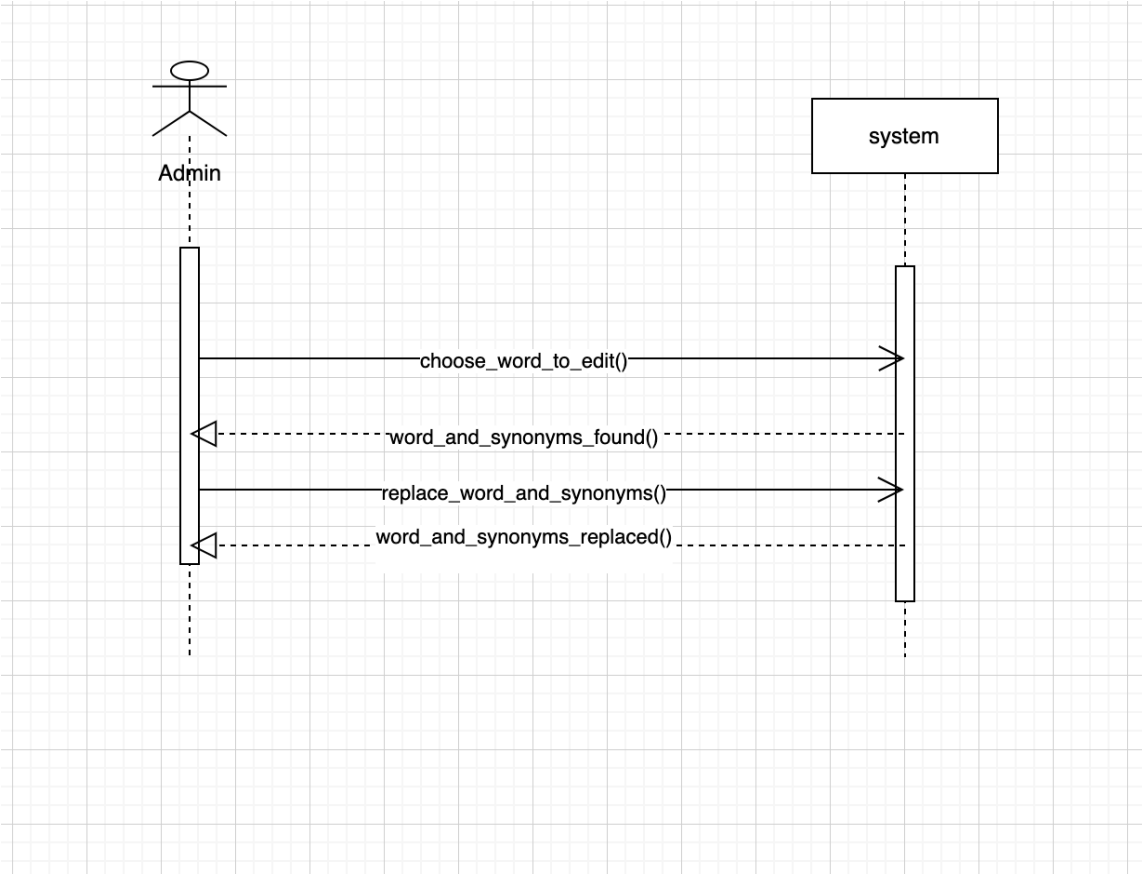
	3. System validates the credentials. 4. Admin can now manage words and synonyms.
Alternative flow	3.a If credentials are incorrect, System displays "incorrect username or password".
Frequency Occurrence	Medium

2.7 System sequence Diagram

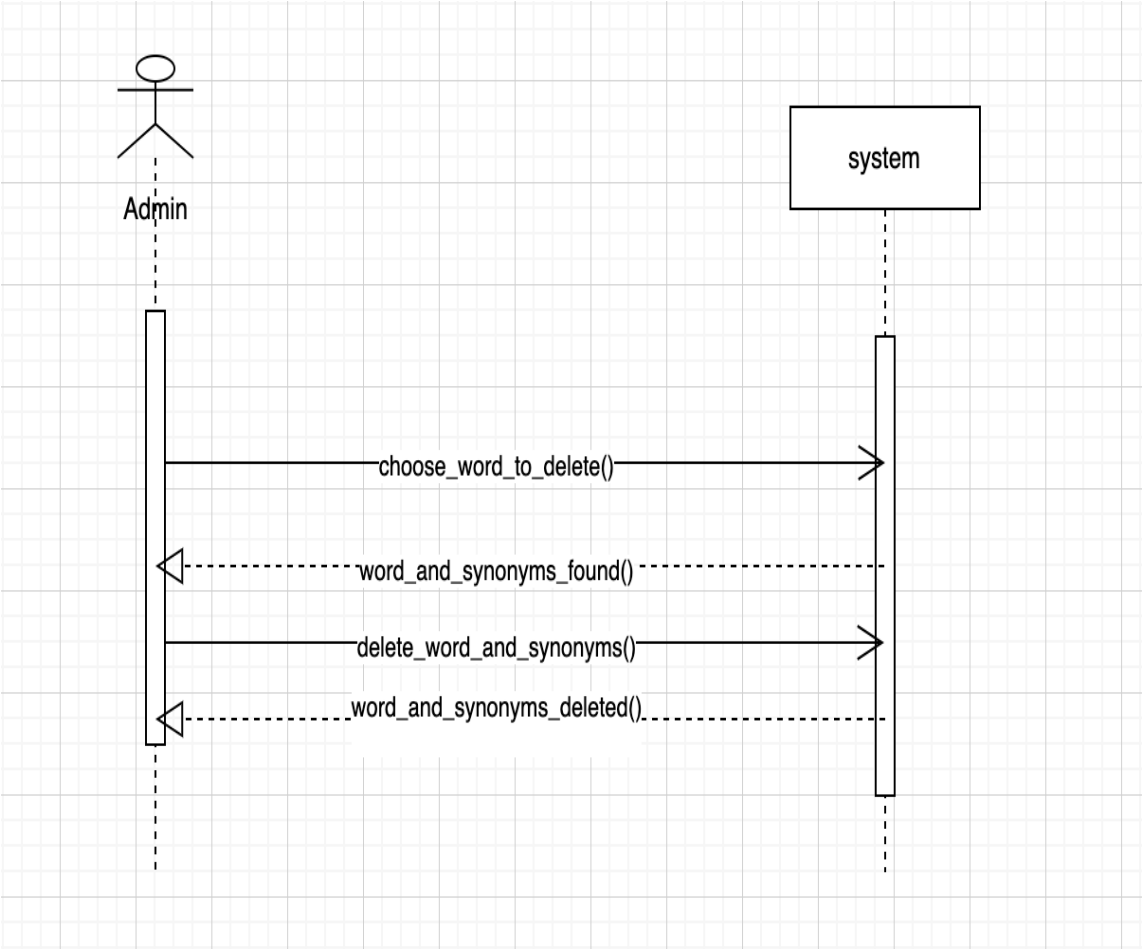
2.7.1 - view synonyms



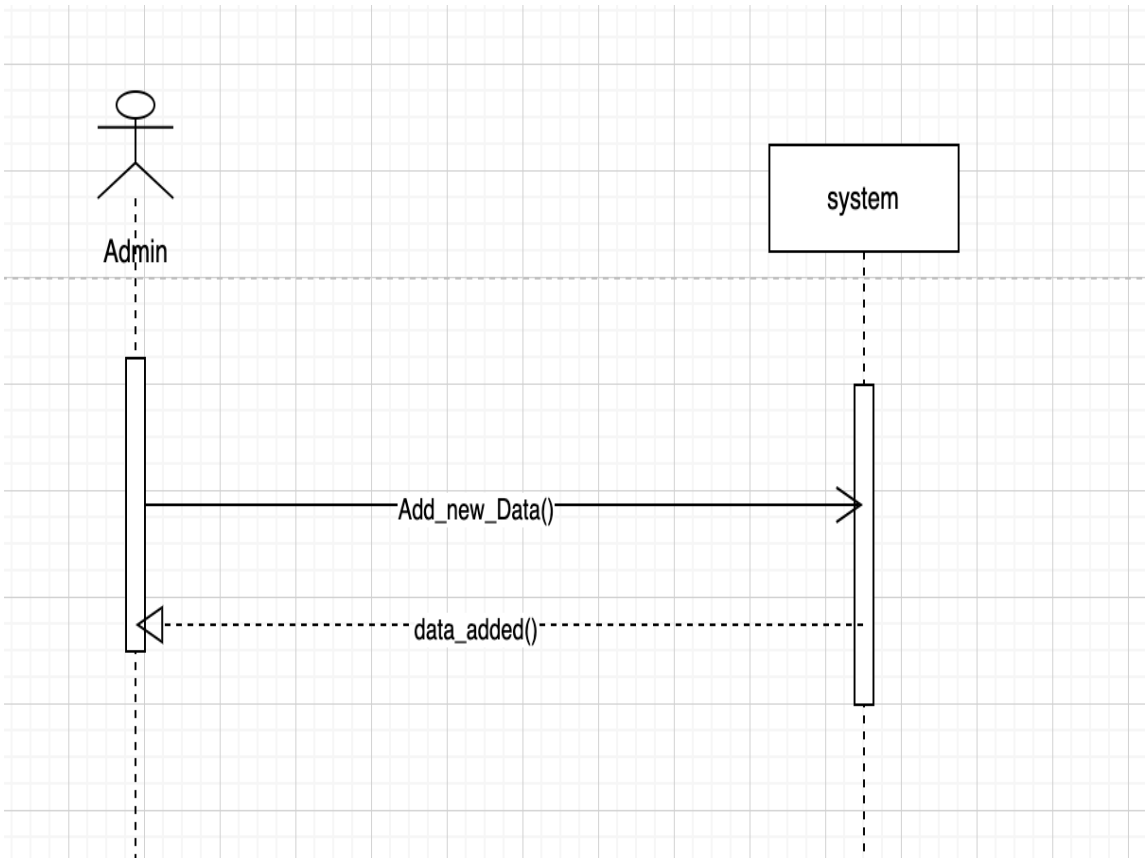
2.7.2 - edit words and synonyms



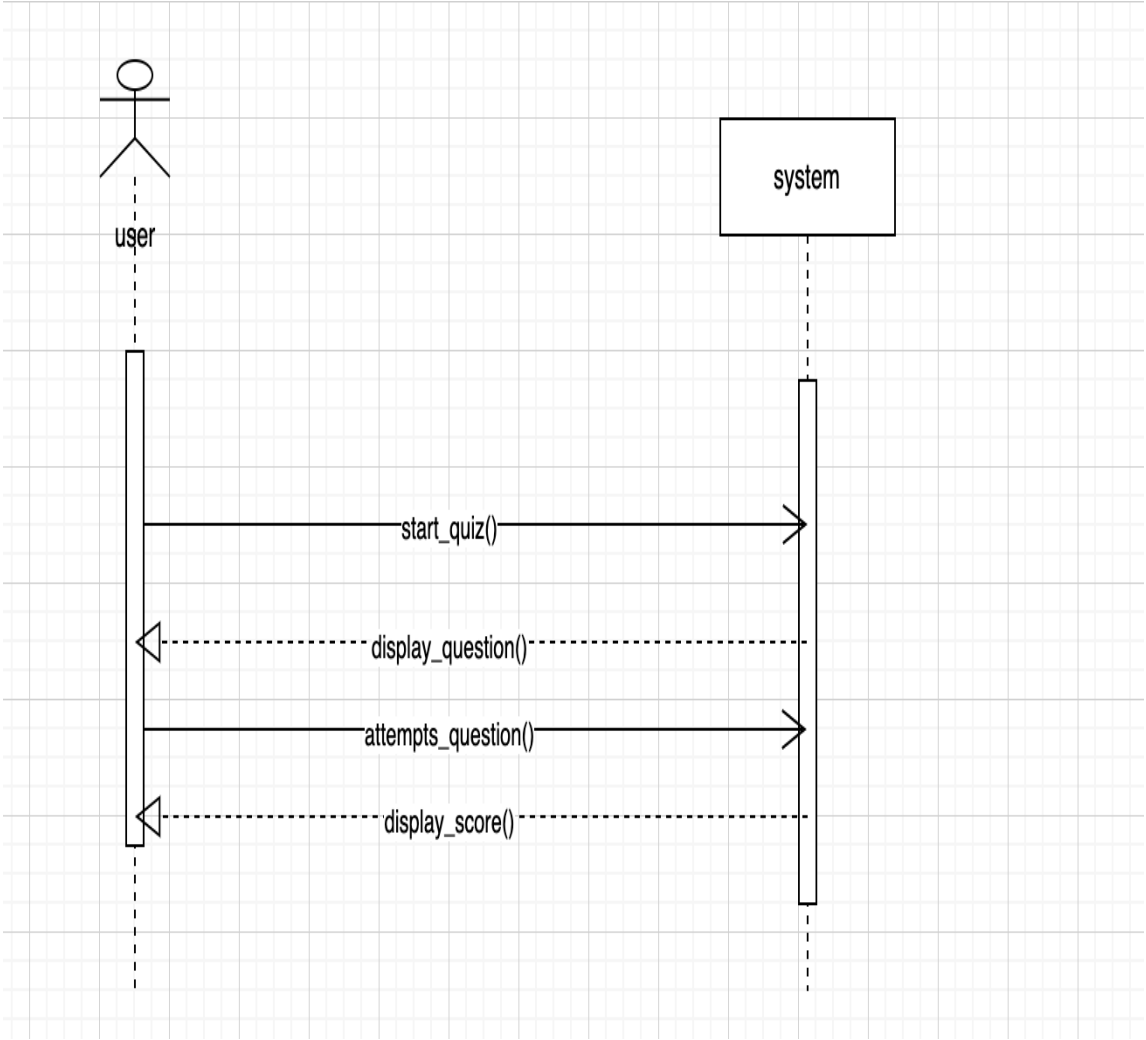
2.7.3 - delete words and synonyms



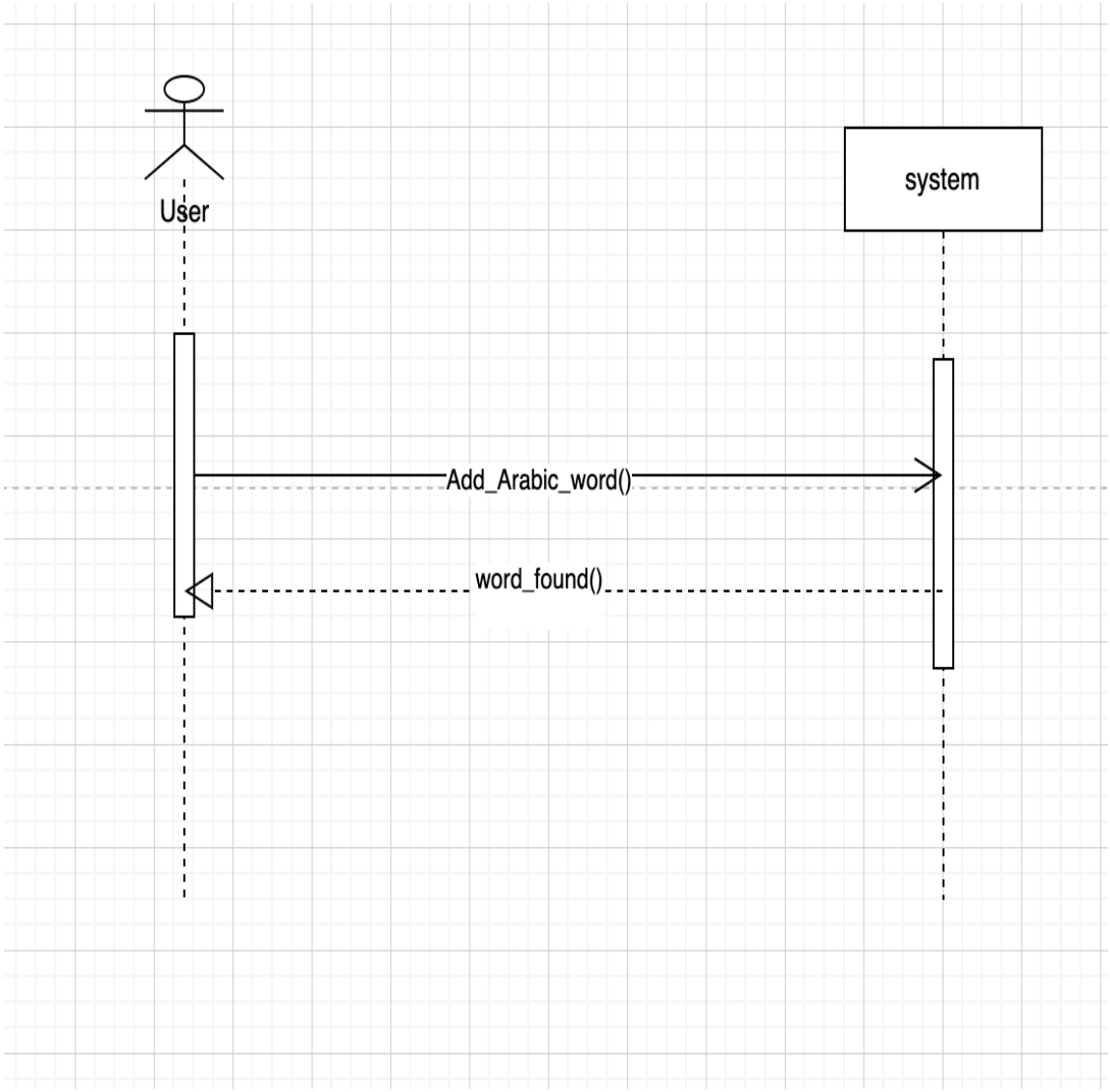
2.7.4 - Add words and synonyms



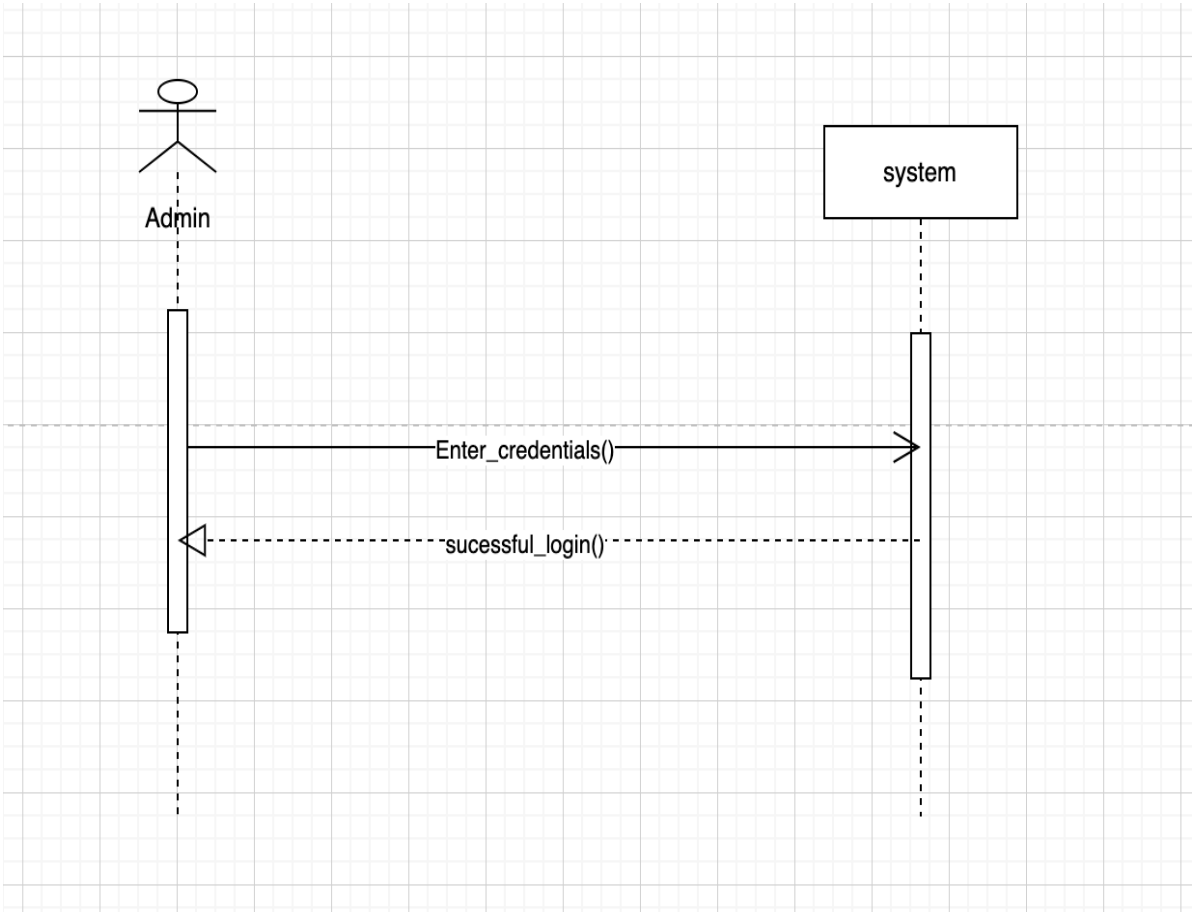
2.7.5 - Take quiz



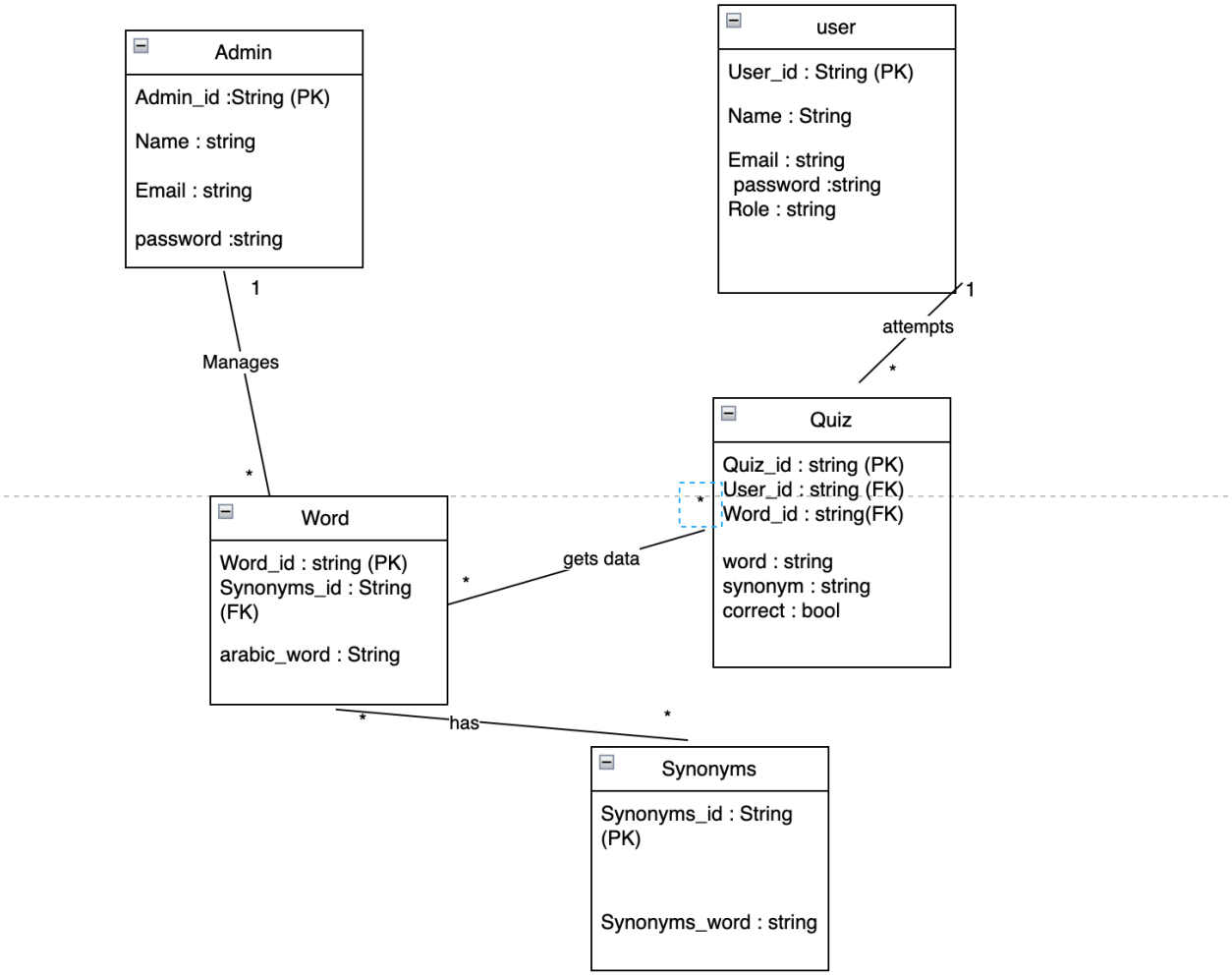
2.7.6 - Search word



2.7.7- login



2.8 ERD (Entity Relationship Diagram)



3. Software Design Description (SDD)

1. INTRODUCTION

1.1 Design Overview

This project is a modular web application developed using Flask (Python) as the backend framework.

- It contains different routes to handle search functionality, admin features, and quizzes.
- The PostgreSQL database is used to store Arabic words and their synonyms.
- The frontend is built using HTML, CSS, and JavaScript, ensuring a responsive and user-friendly interface with proper Arabic text support.

1.2 Requirements Traceability Matrix

Requirment id	Name	Test case	System sequence diagram
Uc 1	Search Word	TC 3.1	Ssd 2.7.6
uc2	View Synonym	TC 3.7	Ssd 2.7.1
Uc 3	Edit word and synonyms	TC 3.6	Ssd 2.7.2
Uc 4	Take Quiz	TC 3.4	Ssd 2.7.5
Uc 5	delete word and synonyms	TC 3.5	Ssd 2.7.3

Uc 6	Add word and synonyms	TC 3.3	Ssd 2.7.4
Uc 7	Login	TC 3.8	Ssd 2.7.7

2. SYSTEM ARCHITECTURAL DESIGN

2.1 Chosen Architecture

The system follows a **client-server architecture**.

- **Frontend (Client)** handles user input, display, and interaction.
- **Backend (Server)** handles logic, data fetching, and admin operations.
- **Database (PostgreSQL)** stores all Arabic words, synonyms, and quiz questions.

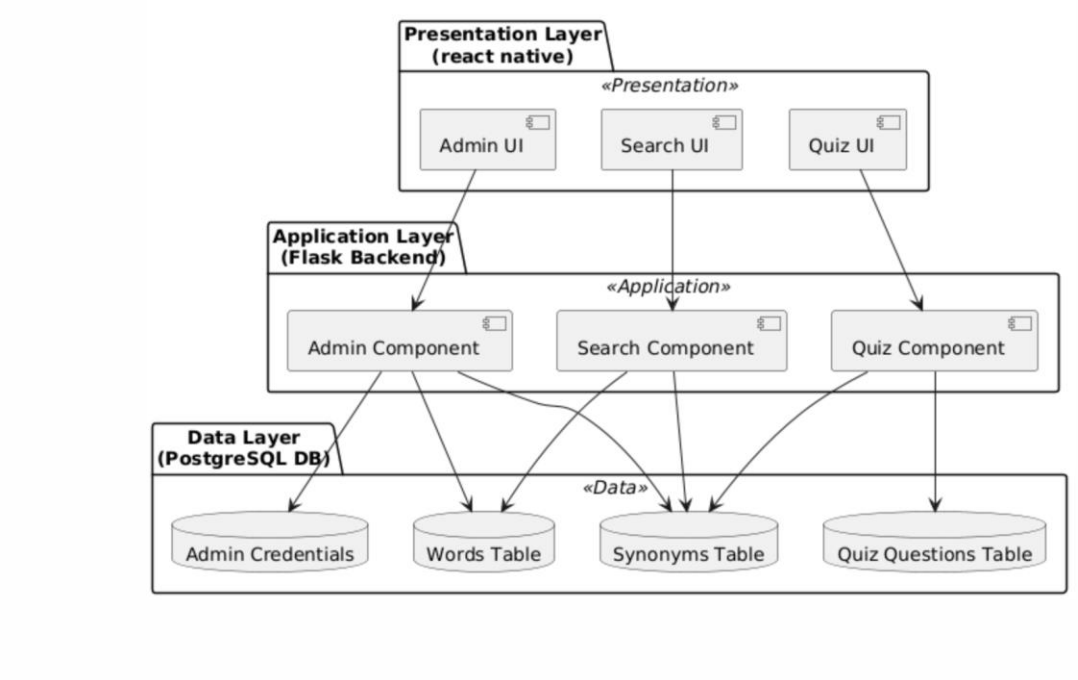
Layered Overview:

- **Presentation Layer**: HTML/CSS/JS (User Interface)
- **Application Layer**: Flask Routes and Logic
- **Data Layer**: PostgreSQL Database

2.2 Alternatives Considered

- **Django**: Considered but found too heavy for this small to medium project.
- **Node.js**: Rejected due to lower built-in support for Arabic string handling and database interaction complexity compared to Python.

2.3 Architectural Diagram



3. DETAILED DESCRIPTION OF COMPONENTS

3.1 Search Component

1. **Input Form**: Accepts Arabic words.

2. **Flask Route**: Search words
3. **Database Interaction**: Queries the word table and retrieves synonyms.

4. **Display**: Returns results to be shown on the frontend.

3.2 Admin Component

1. **Routes**: Admin can add,delete and edit
2. **Access Control**: Only accessible to logged-in admin.

3. **Functionality**:

    - Add: Admin inputs word and synonyms to be saved.

- Edit: Update word or synonym entries.
- Delete: Remove entries.

3.3 Quiz Component

1. **Route**: Quiz
2. **Functionality**:
    - Loads random synonyms from the database.
    - Presents multiple-choice questions.
    - Calculates and displays the score.

4. USER INTERFACE DESIGN

4.1 Description

The system interface is designed to be simple, clean, and Arabic-friendly. It supports proper font size, and input direction.

1. **Search Page**
    - Arabic input field
    - Search button
    - Results panel for synonyms
2. **Admin Panel**
    - Login form
    - Add word/synonym form
    - Edit/Delete options
3. **Quiz Page**

- Displays questions with multiple-choice answers
- Submit button
- Displays result after submission

4. Software Test Documentation (STD)

1. INTRODUCTION

1.1 System Overview

System allows Arabic word search, synonym display, admin management, and quiz.

1.2 Test Approach

- Unit testing (Python)
- Manual UI testing
- DB test queries

2. TEST PLAN

2.1 Features to Be Tested

- Word search
- Synonym retrieval
- Admin word management
- Quiz logic

2.2 Features Not included

- Multi-language support
- Voice input (future version)

2.3 Testing Tools & Environment

- Browser (Chrome/Firefox)
- Postman (for API)
- VSCode Terminal
- SQLite/PostgreSQL for mock DB

3. TEST CASES

3.1 Test Case: Search Existing Word

- **Purpose:** To check if the system correctly returns synonyms for a word that exists.
- **Input:** Arabic word "علم"
- **Expected Output:** List of synonyms related to the word "علم"
- **Pass/Fail Criteria:** The system shows the correct list of synonyms.
- **Procedure:**
    1. Open the search page.
    2. Enter the word "علم".
    3. Click the Search button.
    4. Check if synonyms are displayed.
- **Expected result :** معرفة

3.2 Test Case: Search Non-Existing Word

- **Purpose:** To ensure the system handles words not found in the database.

- **Input:** Word "Taimoor"
- **Expected Output:** Message saying “ no result found “
- **Pass/Fail Criteria:** Message appears if word is not found.

- **Procedure:**

    1. Go to the search page.

    2. Enter “Taimoor” in the input field.

    3. Click Search.

    4. Confirm that the message “ no result found “ is shown.

- **Expected result :** no result found

3.3 Test Case: Add Synonym (Admin)

- **Purpose:** To verify that admin can add a new word and its synonym.

- **Input:** New Arabic word and synonym

- **Expected Output:** Word appears in the database and can be searched

- **Pass/Fail Criteria:** Word and synonym are saved and shown in search results

- **Procedure:**

    1. Admin logs in.

    2. Navigate to the Add Word section.

    3. Enter a new Arabic word and its synonym.

    4. Save the entry.

    5. Use search to verify the word is added successfully.

- **Expected result :** data added successfully

3.4 Test Case: take quiz

- **Purpose:** To check that the quiz feature works and shows correct results.
- **Input:** User answers the quiz questions
- **Expected Output:** Score/result based on the answers
- **Pass/Fail Criteria:** Quiz score is displayed correctly after submission
- **Procedure:**
    1. Click on the Quiz button.
    2. Answer all quiz questions.
    3. Submit the quiz.
    4. Verify that the score is shown based on correct answer.
- **Expected result :** “Score : 7”

3.5 Test Case: Delete Word and Synonym

Purpose: To verify that admin can delete an Arabic word and its synonym from the database.

Input: Arabic word to delete

Expected Output: Word and its synonym no longer appear in search results

Pass/Fail Criteria: Word and synonym are removed successfully from the database

Procedure:

- Admin logs in.
- Go to the "Delete Word" section.
- Select the word and its synonym to delete.

- Confirm deletion.
- Use search to verify the word is no longer available.

Expected Result: Data deleted successfully

3.6 Test Case: Edit Word and Synonym

Purpose: To ensure that admin can update an Arabic word or its synonym.

Input: Existing word/synonym and updated values

Expected Output: Updated word and synonym saved and reflected in search

Pass/Fail Criteria: Changes are saved and appear correctly when searched

Procedure:

- Admin logs in.
- Go to the "Edit Word" section.
- Choose the word/synonym to edit.
- Enter new data.
- Save changes.
- Use search to confirm updates.

Expected Result: Data updated successfully

3.7 Test Case: View Synonyms

Purpose: To verify that users can view synonyms of a given Arabic word.

Input: Arabic word "علم"

Expected Output: List of synonyms for the word

Pass/Fail Criteria: Synonyms appear correctly in results panel

Procedure:

- Open the search page.
- Enter the Arabic word.
- Click on Search.
- View the list of synonyms displayed.

Expected Result: Synonyms displayed successfully

3.8 Test Case: Login

Purpose: To verify that only admins can access secure features through login.

Input: Admin username and password

Expected Output: Admin is logged in and redirected to the admin panel

Pass/Fail Criteria: Admin access granted upon correct credentials

Procedure:

- Open the login page.
- Enter admin username and password.
- Click login.
- Check if admin panel is accessible.

Expected Result: Logged in successfully